

PHYSICAL WORLD

1. How many fundamental forces in nature that govern the diverse phenomena of macroscopic and microscopic world
(1) One (2) Two (3) Three (4) Four
2. The Indian Scientist, who discover 'Inelastic scattering of light by Molecules'
(1) S.N. Bose
(2) S.Chandrasekhar
(3) Homi-Jehangir Bhabha
(4) C.V. Raman
3. Classical physics deals with mainly
(1) Mechanics
(2) Electrodynamics
(3) Optics and thermodynamics
(4) All the above
4. Which of following force is different in nature
(1) Spring force (2) Friction force
(3) Coulomb's force (4) Weight of body
5. In gravity free space, no buoyancy force will act on a body immersed in a fluid as
(1) Buoyancy force is gravitational force
(2) Buoyancy force is electromagnetic force, which arises when pressure of fluid increases with depth due to gravity
(3) When there is no gravity then all other forces are automatically zero
(4) Given statement is wrong, as buoyancy force will act in gravity free spaces
6. If strength of strong nuclear forces is 1 unit then strength of gravitational, weak nuclear and electromagnetic forces will be
(1) 10^{-39} , 10^{-13} , 10^{-2} (2) 10^{-39} , 10^{-2} , 10^{-13}
(3) 10^{-13} , 10^{-39} , 10^{-2} (4) 10^{-2} , 10^{-13} , 10^{-39}
7. In beta decay, if two particles were emitted instead of three then which fundamental law will not hold together
(1) Energy and momentum conservation
(2) Mass and energy conservation
(3) Charge and momentum conservation
(4) Mass and charge conservation
8. As both gravity, electromagnetic forces are long range forces, then in terrestrial mechanism why only gravitational forces are considered ?
(1) Electromagnetic forces are weaker as compared to gravity
(2) Gravitational forces are weaker as compared to electromagnetic forces
(3) Gravitational forces are only attractive while electric forces are repulsive too
(4) Gravitational forces are of longer range
9. Even when strong nuclear forces are strongest forces in nature, why do we not observe it in daily phenomenon ?
(1) They are long range forces
(2) They are short range force which act with in atom
(3) They are very short range force which act with in nucleus
(4) They are balanced by other forces